MODULE 3 FASCICLES AND LEVERS

Fascicle Arrangement & Lever Systems.

MUSCLE ARRANGEMENT OF FASCICLES AND LEVER SYSTEMS

A reference for the fascicle arrangement and lever systems video and lab. This chapter covers how the fascicles of a muscle are arranged, how muscles are named, how they work in groups, and the bone-and-joint lever systems they act on.

BY THE END

1. Identify the patterns of fascicle arrangement and give an example muscle for each.
2. Explain how fascicle architecture trades range of motion against power.
3. Name the criteria used to name muscles, and classify the roles muscles play in a movement.
4. Identify the parts of a lever and classify first, second, and third class levers in the body.

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: Fascicle arrangement and levers

drsrennie-stack.github.io/new-build-bio4-solano/muscle-fascicles-concept-videos.html

From Fascicles to Movement, an Overview

A skeletal muscle moves a bone by pulling across a joint. How its fascicles are arranged, what it is named, the team it works in, and the lever it acts on all shape the movement it produces.

THE STARTING VOCABULARY

Structure	What it is
Fascicle	A bundle of muscle fibers wrapped in perimysium, the unit whose arrangement gives a muscle its shape
Origin	The attachment of a muscle to the more stationary bone
Insertion	The attachment of a muscle to the more movable bone, pulled toward the origin
Action	The movement a muscle produces when it contracts and pulls the insertion toward the origin

Fascicle Arrangement Patterns

The fascicles of a muscle run in a set pattern, and that pattern gives the muscle its overall shape. Compare the patterns by how the fascicles lie and an example muscle.

PATTERN, ARRANGEMENT, AND EXAMPLE

Pattern	Fascicle arrangement	Example muscle
Parallel	Fascicles run parallel to the long axis of the muscle	Sartorius, rectus abdominis
Fusiform	Fascicles run parallel, but the muscle has a thick central belly that tapers to a tendon at each end	Biceps brachii
Circular	Fascicles arranged in concentric rings around an opening, forming a sphincter	Orbicularis oris, orbicularis oculi
Convergent	A broad origin whose fascicles converge onto a single tendon, giving a triangular or fan shape	Pectoralis major
Pennate	Short fascicles attach at an angle to a central tendon, like the barbs of a feather. Unipennate has fascicles on one side, bipennate on both, multipennate on a branched tendon	Unipennate, extensor digitorum; bipennate, rectus femoris; multipennate, deltoid

Architecture, Power, and Range

Fascicle architecture is a trade-off. A muscle can be built for a wide range of motion or for power, but not for both at once. Compare the two extremes.

THE TRADE-OFF

Architecture	Fascicle length and number	Range of motion	Power
Parallel and fusiform	Long fascicles, fewer of them per unit of muscle	Larger range of motion, the muscle shortens by a greater distance	Less power, fewer fibers pull on the tendon
Pennate	Short fascicles, many of them packed at an angle per unit of muscle	Smaller range of motion, the angled fascicles shorten by a shorter distance	Greater power, many more fibers pull on the tendon

WHY THE TRADE-OFF EXISTS

Structure	What it is
Range of shortening	A muscle fiber can shorten to roughly two thirds of its resting length, so a muscle with longer fascicles moves its insertion farther

How Muscles Are Named

A muscle's name is a description. Most names combine two or more of these criteria, so the name itself tells you where the muscle is, what it looks like, or what it does.

THE NAMING CRITERIA

Naming criterion	What it describes	Examples
Direction of fascicles	How the fascicles run relative to the body's midline	rectus, straight; oblique, at an angle; transversus, across
Relative size	The size of the muscle compared with its neighbors	maximus, minimus, longus, brevis
Shape	The overall geometric shape of the muscle	deltoid, triangular; trapezius, trapezoid; rhomboid
Location	The bone or body region the muscle is near	tibialis anterior, frontalis, intercostals
Number of origins	How many heads, or origin tendons, the muscle has	biceps, two; triceps, three; quadriceps, four
Origin and insertion	The muscle's two attachment points	sternocleidomastoid, from the sternum and clavicle to the mastoid process
Action	The movement the muscle produces	flexor, extensor, abductor, adductor, levator

Muscles Working in Groups

Almost no movement is the work of one muscle. Muscles act in teams, and the same muscle can play different roles in different movements.

THE FOUR ROLES

Role	Function in the movement	Example
Agonist	The prime mover, the muscle chiefly responsible for producing a given movement	Brachialis during elbow flexion
Antagonist	Opposes the agonist, relaxing and lengthening as the agonist contracts; it can slow or reverse the movement	Triceps brachii during elbow flexion
Synergist	Assists the agonist, adding force or preventing an unwanted movement	Biceps brachii assisting elbow flexion
Fixator	A synergist that holds the origin bone steady so the agonist can act efficiently on the insertion bone	Scapular muscles steadying the scapula during arm movements

Lever Systems

When a muscle pulls on a bone, the bone acts as a lever and the joint as its pivot. Every body movement is a lever at work.

THE PARTS OF A LEVER

Structure	What it is
Lever	A rigid bar that moves around a fixed point; in the body, a bone is the lever
Fulcrum	The fixed point the lever turns around; in the body, a joint is the fulcrum
Effort	The force applied to move the lever; in the body, the pull of a contracting muscle
Load	The resistance moved by the lever; in the body, the weight of the body part plus anything it carries
Mechanical advantage	When the effort acts farther from the fulcrum than the load, so a small effort moves a large load
Mechanical disadvantage	When the effort acts closer to the fulcrum than the load, so a large effort is needed, but the load moves fast and far

The three classes of lever

Levers are sorted into three classes by the order of the fulcrum, effort, and load along the bar.

ARRANGEMENT AND AN EXAMPLE IN THE BODY

Lever class	Arrangement along the lever	Example in the body
First class lever	The fulcrum lies between the effort and the load; think of a seesaw	Nodding the head: the posterior neck muscles, the atlanto-occipital joint, and the weight of the face
Second class lever	The load lies between the fulcrum and the effort; think of a wheelbarrow	Standing on tiptoe: the ball of the foot, the body's weight, and the calf muscles pulling up the heel
Third class lever	The effort lies between the fulcrum and the load; think of tweezers. The most common lever in the body	Flexing the elbow: the elbow joint, the biceps pull, and the weight held in the hand

KEY POINT

Most body movements run on third class levers, so most muscles work at a mechanical disadvantage. The body trades force for speed and range, which is why a muscle has to pull far harder than the weight it is lifting.

ARCHITECTURE READS OFF THE BODY

Fascicle architecture is visible in the clinic. A **pennate** muscle like the deltoid packs many short fascicles around a tendon for power, while a **parallel** muscle like the sartorius trades power for a long, sweeping range. A surgeon planning a tendon transfer matches the architecture to the job the muscle must take over.

Muscle names are a working vocabulary. Reading flexor carpi radialis tells a clinician the action (flexor), the region (carpi, the wrist), and the position (radialis, the radial side) before the muscle is ever seen, which is why the naming criteria are worth learning as a system.

Most body movements run on **third class levers**, which work at a mechanical disadvantage: the muscle must generate a large force, but it buys speed and a wide range of motion. That trade-off is why a small change in where a tendon inserts can noticeably change a joint's strength and speed.